

Financial Time Series Analysis

Lecture 11

- ▶ TA Session: Friday, April 30, 11:00am-12:00pm
- ▶ TA Session: Monday, May 3, 7:30-8:30pm

Outline

- ▶ Final Exam, Monday, May 10, 10:00am (EDT)
- ▶ Other GARCH Models
- ▶ Multiple Time Series
 - ▶ Basic Details
 - ▶ VAR Processes (Vector Autoregressive Processes)
- ▶ Cointegration

Nature of the Final Exam: 1

- ▶ Final exam is comprehensive, from stylized facts, stationarity and MDS to state space models (lecture 13).
- ▶ Exam length will be 1.5 or 2 hours, partly multiple choice, partly calculations (like homework calculation problems).
- ▶ Administered under ProctorTrack and Canvas quiz with an open zoom link.
- ▶ Closed book but one 8.5x11 (two-sided) cheat sheet) is permitted. No other computational or communication devices except to download and upload the exam.

Nature of the Final Exam: 2

- ▶ The test **is not** intended to be a test of memory, especially the exact use of time series commands like: `statsmodels.tsa.stattools.adfuller`
- ▶ Rather the test is intended to see whether you have an understanding of the time series concepts, models and methods and their appropriate use.
- ▶ I will post a previous exam next week to help you study for the exam.
- ▶ The basis for the final will be the lecture slides and the homework problems. The Course Notes also provide a study guide.

Other Volatility Models: 1

- ▶ There are many variations on the GARCH model to address shortcomings and extend its applicability. We mention just a few of the more common ones.
- ▶ Consider **GARCH-M (GARCH in the mean)**. Normally, we have a mean model and a separate volatility model.
- ▶ The GARCH-M model includes a volatility term in the mean equation. This can arise because it might be appropriate to include a risk premium associated with volatility in the returns part of the model (see homework problem 5.1).

Other Volatility Models: 2

- ▶ This model can be expressed as

$$X_t = \mu + c\sigma_t^2 + a_t,$$

$$a_t = \epsilon_t \sigma_t,$$

$$\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

- ▶ Here μ and c are constant. If $c > 0$ then the return process is positively related to the past volatility. Other specifications of the volatility term have been used including $c\sigma_t$ and $c \log(\sigma_t^2)$.

Other Volatility Models: 3

- ▶ One notable shortcoming of the GARCH model is the presence of volatility terms being squared ($a_{t-1}^2, \sigma_{t-1}^2$)
- ▶ This means that both positive and negative changes are treated the same in updating the volatility equation.
- ▶ However, one of the stylized facts is that the relationship should be asymmetric, with larger increases in volatility coming from negative changes than come from positive changes. The **EGARCH (Exponential GARCH)** model addresses this shortcoming.
- ▶ This was introduced in 1991 by Nelson. He considered weighting the innovations:

Other Volatility Models: 4

- ▶ Nelson began with the $\{\epsilon_t, t \geq 0\}$ innovations and weighted them $\{g(\epsilon_t), t \geq 0\}$ where

$$g(\epsilon_t) = \theta\epsilon_t + \gamma[|\epsilon_t| - E(|\epsilon_t|)],$$

where θ and γ are constants. Both components are iid with mean 0. The asymmetry can be seen by writing out the two cases:

$$g(\epsilon_t) = \begin{cases} (\theta + \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t \geq 0, \\ (\theta - \gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t < 0, \end{cases}$$

For the leverage effect we would expect to have $\theta < 0$ and $\gamma > 0$.

Other Volatility Models: 5

The name, E-GARCH, comes from the modeling of σ_t :

$$a_t = \sigma_t \epsilon_t, \{\epsilon_t\}, i.i.d., E(\epsilon_t) = 0,$$

$$\log(\sigma_t^2) = \omega + \sum_{i=1}^p \beta_i \log(\sigma_{t-i}^2) + \sum_{j=1}^q g_j(\epsilon_{t-j}),$$

$$g_j(\epsilon_t) = \alpha_j \epsilon_t + \gamma_j (|\epsilon_t| - E(|\epsilon_t|)).$$

- ▶ The model can retain the good properties of the GARCH model and in addition add the leverage effect.
- ▶ The Python package ARCH includes a number of variations on the GARCH model including E-GARCH, FIGARCH, and APARCH among others.

Multivariate Time Series: 1

- ▶ Many of the ideas we have studied for univariate time series carry over to the multivariate case. This is important because we would like to conduct an analysis that not only deals with the individual aspects of some asset but its relationship with other assets.
- ▶ Unfortunately, as we go to the multivariate case, the number of parameters can skyrocket - parameters for the individual components and parameters that deal with dependency relationships.
- ▶ We begin with a m -dimensional time series $\mathbf{X}_t = (X_{1t}, \dots, X_{mt})$.

Multivariate Time Series: 2

- ▶ As each component is a time series, each will have a mean function μ_{it} and an autocovariance function, $\Gamma_{ii}(t+h, t) = \text{Cov}(X_{i(t+h)}, X_{it}) = E(X_{i(t+h)}X_{it}) - \mu_{it}^2$.
- ▶ The covariance function (now a matrix) is given by

$$\mathbf{\Gamma}(t+h, t) = E((\mathbf{X}_{t+h} - \boldsymbol{\mu}_{t+h})(\mathbf{X}_t - \boldsymbol{\mu}_t)^T).$$

- ▶ We can define a weakly stationary multivariate time series by requiring that the first two moments, the mean vector, $\boldsymbol{\mu}_t$ and the autocovariance matrix, $\mathbf{\Gamma}(t+h, t)$, are independent of t .
- ▶ The diagonal elements of the $\mathbf{\Gamma}(h)$, $\gamma_{ii}(h)$, are the autocovariance functions of the individual component time series.

Multivariate Time Series: 3

- ▶ We can also create the autocorrelation matrix, $\mathbf{R}(h) = \rho_{ij}(h)$, and a typical entry can be computed by

$$\rho_{ij}(h) = \frac{\gamma_{ij}(h)}{(\gamma_{ii}(0)\gamma_{jj}(0))^{1/2}}.$$

- ▶ From these definitions, we can define a vector white noise process.

Definition

$\{\epsilon_t, t \geq 0\}$ is $\text{WN}(\mathbf{0}, \mathbf{\Sigma})$ if and only if

- ▶ The ϵ_t process is weakly stationary,
- ▶ $\mu = \mathbf{0}$,
- ▶

$$\Gamma(h) = \begin{cases} \mathbf{\Sigma} & \text{if } h = 0, \\ \mathbf{0} & \text{if } h \neq 0. \end{cases}$$

Multivariate Time Series: 4

Note that $\mathbf{R} = (R_{ij})$.

- ▶ $R_{ii}(h)$ is the autocorrelation function of X_{it} .
- ▶ $R_{ij}(h) = \rho(X_{i(t+h)}, X_{jt})$ is called the **cross-correlation function (CCF)** of $\{X_{it}\}$ and $\{X_{jt}\}$
- ▶ Note that $R_{ij}(h) = \rho(X_{i(t+h)}, X_{jt}) \neq \rho(X_{it}, X_{j(t+h)})$.
- ▶ Rather $R_{ij}(-h) = \rho(X_{it}, X_{j(t+h)}) = \rho(X_{j(t+h)}, X_{it}) = R_{ji}(h)$
- ▶ In matrix form, $\mathbf{R}(h) = \mathbf{R}(-h)^T$.
- ▶ The sample estimates for the first and second moments are:

Multivariate Time Series: 5

The usual estimate of the common mean of $\mathbf{X}$, $E(\mathbf{X}) = \boldsymbol{\mu} = (\mu_1, \dots, \mu_m)$ is the sample mean,

$$\bar{\mathbf{X}}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i.$$

- ▶ Under the assumption that the multivariate time series is weakly stationary, then the sample mean, $\bar{\mathbf{X}}_n$, converges to $\boldsymbol{\mu}$ and is asymptotically normal.
- ▶ That is, $\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu})$ converges weakly to a $N_d(\mathbf{0}, \boldsymbol{\Sigma}^*)$ where $\boldsymbol{\Sigma}^*$ depends upon the properties of the underlying white noise process.

Multivariate Time Series: 6

- ▶ We next turn to the estimation:

$$\boldsymbol{\Gamma}(h) = E((\mathbf{X}_{t+h} - \boldsymbol{\mu})(\mathbf{X}_t - \boldsymbol{\mu})^T),$$

- ▶ The obvious sample estimate is

$$\hat{\boldsymbol{\Gamma}}(h) = \frac{1}{n} \sum_{i=1}^{n-h} (\mathbf{X}_{t+h} - \bar{\mathbf{X}}_n)(\mathbf{X}_t - \bar{\mathbf{X}}_n)^T, \quad 0 \leq h \leq n-1.$$

- ▶ Letting $\hat{\boldsymbol{\Gamma}}(h) = (\hat{\gamma}_{ij})$ and $\hat{\mathbf{R}}(h) = (\hat{\rho}_{ij})$, we have



$$\hat{\rho}_{ij}(h) = \frac{\hat{\gamma}_{ij}(h)}{(\hat{\gamma}_{ii}(0)\hat{\gamma}_{jj}(0))^{1/2}}.$$

Vector Autoregressive Processes

- ▶ We want to deal with the simultaneous behavior of financial time series in pairs trading and portfolio construction.
- ▶ The generalization of univariate AR processes to multivariate AR processes is straightforward. These are called **Vector Autoregressive Processes** or VAR(p).
- ▶ Let $\mathbf{X}_t = (X_{1t}, \dots, X_{mt})$ be an m-dimensional vector time series satisfying

$$\mathbf{X}_t = \mathbf{a}_0 + \sum_{i=1}^p \mathbf{A}_i \mathbf{X}_{t-i} + \boldsymbol{\epsilon}_t,$$

where $\mathbf{a}_0$ is an m-dimensional vector, $\mathbf{A}_1, \dots, \mathbf{A}_p$ are $m \times m$ matrices and $\{\boldsymbol{\epsilon}_t\}$ are i.i.d. m-dimensional random vectors with mean $\mathbf{0}$ and some covariance matrix $\boldsymbol{\Gamma}$.

- ▶ For stationarity, we would want each of the individual series to be stationary as well as the vector process.

Stationarity of VAR(1) Processes

- ▶ Suppose we consider the VAR(1) process

$$\mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} + \epsilon_t.$$

- ▶ One can calculate $\Sigma_{\mathbf{X}_t}$ by iteration. In the simple case with $\mathbf{a} = 0$ and $\Gamma = \sigma^2 \mathbf{I}$, then as $t \rightarrow \infty$

$$\Sigma_{\mathbf{X}_t} \rightarrow \sigma^2 (\mathbf{I} - \mathbf{A}\mathbf{A}^T)^{-1}.$$

- ▶ This is the multivariate analog of the univariate result in which

$$\text{Var}(X_t) \rightarrow \frac{\sigma^2}{1 - \beta^2}.$$

- ▶ Need the roots of the equation $\det(\mathbf{I} - \mathbf{A}x) = 0$ to have modulus larger than 1, in analogy with the AR(1) process.

Stationarity of VAR(p) processes

- ▶ For the VAR(p) processes, the stationarity condition is similar to that for AR(p). Consider the roots of the polynomial equation

$$\det\left(I - \sum_{i=1}^p \mathbf{A}_i x^i\right) = 0.$$

- ▶ The VAR(p) process is stationary if and only if the roots of the above equation have modulus strictly larger than 1.

Cointegration: 1

In the introduction to their 1987 paper, Engle and Granger wrote

An individual economic variable, viewed as a time series, can wander extensively and yet some pairs of series may be expected to move so that they do not drift too far apart. Typically economic theory will propose forces which tend to keep such series together. Examples might be short and long term interest rates, capital appropriations and expenditures, household income and expenditures, and prices of the same commodity in different markets or close substitutes in the same market.

Cointegration: 2

- ▶ In financial markets, the variables in question are stock prices. No single stock price can be expected to be stationary but will vary, potentially widely.
- ▶ The crucial idea is to find a subset of these processes that “move together” in the sense that some linear combination is a stationary process.
- ▶ If two stock price processes are “cointegrated,” then over the long run these two processes are “tied together” and make a good candidate for a trading strategy.
- ▶ This strategy, known as “pairs trading” would track two stock price paths (denominated in standard units). We expect to see mean reversion.

Cointegration: 3

- ▶ If one of the stocks' price rises above the other, when the difference is sufficiently large, short the “overpriced” stock and long the “underpriced” stock. Close the trade when they revert to their mean.
- ▶ This strategy is considered to be zero cost and market neutral.
- ▶ Zero cost because your portfolio is long and short the same amount of money (although there may be extra costs in shorting stocks and other technical difficulties).
- ▶ The strategy is considered to be market neutral. As long as the cointegration relationship holds, it does not matter if markets move up or down, as this pair is moving together, and the profit comes from collecting the spread at the time the trade is initiated.

Gatev et al: Pairs Trading

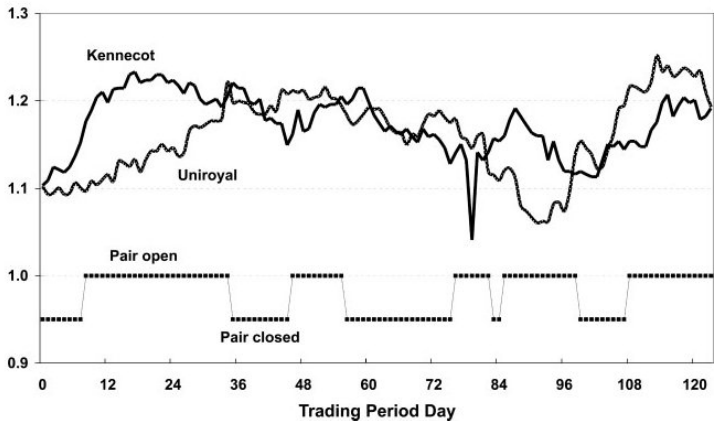


Figure 1
Daily normalized prices: Kennecott and Uniroyal (pair 5)
Trading period August 1963–January 1964.

Cointegration: 4

- ▶ Definition: The components of $\mathbf{X}$ are said to be **cointegrated of order d, b** (denoted $CI(d, b)$) if:
 - a) All components of $\mathbf{X}_t$ are $\mathcal{I}(d)$,
 - b) There exists a vector $\alpha \neq \mathbf{0}$ for which if $Z_t = \alpha \mathbf{X}_t$ is $\mathcal{I}(d - b)$ with $b > 0$.
- ▶ The vector α is called a **cointegrating vector**.

Cointegration: 5

- ▶ Recall that 2 or more time series are said to be **cointegrated** if some linear combination is stationary.
- ▶ Key Questions:
 - ▶ Is cointegration the same as correlation?
 - ▶ How to determine if two or more series are cointegrated?
 - ▶ How to find a “cointegrating vector?”
 - ▶ How to find cointegrated series in a large universe of stocks?

Cointegration: 6

- ▶ Consider three time series, X_t, Y_t, W_t and three independent sequences of i.i.d. random variables, $\{\epsilon_{X,t}\}$, $\{\epsilon_{Y,t}\}$, and $\{\epsilon_t\}$.
- ▶ Define

$$W_t = W_{t-1} + \epsilon_t,$$

$$X_t = W_t + \epsilon_{X,t},$$

$$Y_t = W_t + \epsilon_{Y,t},$$

- ▶ The X_t and Y_t are simply noisy versions of W_t , and W_t is a random walk.
- ▶ Note that $\nabla X_t = W_t + \epsilon_{X,t} - W_{t-1} - \epsilon_{X,t-1} = \epsilon_{X,t} - \epsilon_{X,t-1} + \epsilon_t$ which is stationary (a linear combination of stationary processes).
- ▶ It follows that X_t is $\mathcal{I}(1)$.

Cointegration: 7

- ▶ Similarly, $\nabla Y_t = W_t + \epsilon_{Y,t} - W_{t-1} - \epsilon_{Y,t-1} = \epsilon_{Y,t} - \epsilon_{Y,t-1} + \epsilon_t$ which is stationary.
- ▶ It follows that Y_t is $\mathcal{I}(1)$.
- ▶ X_t and Y_t are cointegrated since both are $\mathcal{I}(1)$ and $X_t - Y_t = \epsilon_{X,t} - \epsilon_{Y,t}$, which is a stationary noise process with cointegrating vector $(1, -1)$.
- ▶ Even though these two processes are cointegrated, they may have very low correlation when $\text{Var}(\epsilon_{X,t}) = \sigma_X^2$ and $\text{Var}(\epsilon_{Y,t}) = \sigma_Y^2$ are large compared with $\text{Var}(\epsilon_t) = \sigma^2$.

$$\text{Var}(X_t) = \text{Var}(W_t + \epsilon_X) = t\sigma^2 + \sigma_X^2,$$

$$\text{Var}(Y_t) = \text{Var}(W_t + \epsilon_Y) = t\sigma^2 + \sigma_Y^2.$$

Cointegration: 8

$$\begin{aligned}\text{Cov}(X_t, Y_t) &= \text{Cov}(W_t + \epsilon_{X,t}, W_t + \epsilon_{Y,t}) \\ &= \text{Cov}(W_t, W_t) \\ &= t\sigma^2\end{aligned}$$

The correlation between X_t and Y_t is given by

$$t\sigma^2 / \sqrt{(t\sigma^2 + \sigma_X^2)(t\sigma^2 + \sigma_Y^2)},$$

which gives

$$1 / \sqrt{(1 + \frac{\sigma_X^2}{t\sigma^2})(1 + \frac{\sigma_Y^2}{t\sigma^2})},$$

Cointegration: 9

- ▶ If one or both of σ_X^2 and σ_Y^2 is large compared with $t\sigma^2$, then the denominator can be large compared with the numerator (1).
- ▶ Thus, even though the X_t and Y_t are cointegrated, their correlation can be small.
- ▶ Nevertheless, as t increases, this ratio is decrease and the correlation between X_t and Y_t will increase to 1. We can, however, modify this example

$$W_t = W_0 + \epsilon_t,$$

$$X_t = W_t + \epsilon_{X,t},$$

$$Y_t = W_t + \epsilon_{Y,t},$$

Here, rather than performing a random walk, W_t randomly moves about a fixed point W_0 .

Cointegration: 10

- ▶ With this modification, the quantity $t\sigma^2$ becomes σ^2 , and if either or both of σ_X and σ_Y is large compared with σ^2 , the correlation between X_t and Y_t will be near 0.
- ▶ When two time series, X_1 and X_2 are cointegrated, there will be a unique cointegrating vector of the form $(1, \alpha)$.
- ▶ The uniqueness holds since if there were two such vectors, $(1, \alpha_1)$ and $(1, \alpha_2)$ then $(X_1 + \alpha_1 X_2) - (X_1 + \alpha_2 X_2) = (\alpha_1 - \alpha_2)X_2$ would be stationary, in contradiction to the assumption that X_2 is not stationary.
- ▶ This uniqueness fails when $m > 2$. There may be as many as $m - 1$ linearly indep. cointegrating vectors,
- ▶ Those cointegrating vectors will span the cointegrating space, but the choice of vectors is not unique,

Cointegration: 11

- ▶ Cointegration does not imply correlation. The reverse is also true. Series that are highly correlated are not necessarily cointegrated. Consider three time series, $\{X_t\}$, $\{Y_t\}$ and $\{Z_t\}$ with

$$X_t = X_{t-1} + \epsilon_{X,t},$$

$$Y_t = Y_{t-1} + \epsilon_{Y,t},$$

$$Z_t = X_t + Y_t,$$

where $\{\epsilon_{X,t}\}$ and $\{\epsilon_{Y,t}\}$ are iid mean 0 processes independent of each other.

- ▶ $\{X_t\}$ and $\{Y_t\}$ are independent random walks whose innovations have variance σ_X^2 and σ_Y^2 while $\{Z_t\}$ the sum of these two random walks.

Cointegration: 12

- ▶ X_t and Z_t are not cointegrated. Although each is $\mathcal{I}(1)$ (as is Y_t), any linear combination of the two will involve both X_t and Y_t , neither of which is stationary.
- ▶ $\text{Cov}(X_t, Z_t) = \text{Var}(X_t) = t\sigma_X^2$
- ▶ $\text{Var}(X_t) = t\sigma_X^2$ and $\text{Var}(Z_t) = t\sigma_X^2 + t\sigma_Y^2$
- ▶ $\rho_{X_t, Z_t} = 1/\sqrt{1 + (\sigma_Y^2/\sigma_X^2)}$,
- ▶ If σ_Y/σ_X is small, then the correlation will be large. Hence this is an example of two series that are not cointegrated but can have very high correlation.

Cointegration: 13

- ▶ Both $\{X_t\}$ and $\{Y_t\}$ are $\mathcal{I}(1)$ by direct differencing.
- ▶ Every linear combination of $\{X_t\}$ and $\{Y_t\}$ is also $\mathcal{I}(1)$, hence they are not cointegrated.
- ▶ The squared correlation coefficient is $1/(1 + \frac{\sigma_Y^2}{\sigma_X^2})$. Thus if σ_X is small compared with σ_Y , then the correlation will be near 1.
- ▶ Thus this pair can be highly correlated but is not cointegrated.

Cointegration: 14

- ▶ There are two major approaches to determining the presence of cointegration of two or more time series:

1. Granger-Engle
2. Johansen

- ▶ **Granger-Engle Method**

Represent one time series as a linear combination of the others:

$$X_{jt} = \sum_{i \neq j} \beta_i X_{it} + \epsilon_t.$$

- ▶ Fit using least squares and test whether the residuals are stationary using Dickey-Fuller
- ▶ If the residuals are stationary, then the cointegrating vector is $(\beta_1, \dots, \beta_{j-1}, \mathbf{1}, \beta_{j+1}, \dots, \beta_m)$. There can be several cointegrating vectors.

Cointegration: Johansen 1

- ▶ We begin with a VAR(1) process and later generalize to an AR(p) process.
- ▶ We assume that the process is $\mathcal{I}(1)$. If it is $\mathcal{I}(0)$, then the original series are all stationary, and there is no substance to cointegration.
- ▶ If the process is $\mathcal{I}(k)$, $k \geq 2$, then we difference $k - 1$ times to bring it back to $\mathcal{I}(1)$. Let

$$\mathbf{X}_t = \mathbf{a} + \mathbf{A}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t.$$

- ▶ Subtract $\mathbf{X}_{t-1}$ from both sides to obtain

$$\nabla \mathbf{X}_t = \mathbf{a} + (\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t.$$

Cointegration: Johansen 2

- ▶ Now if each component of $\mathbf{X}$ is $\mathcal{I}(1)$, then $\nabla \mathbf{X}$ is stationary. Thus the right side of the equation must also be stationary.
- ▶ But $(\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1}$ is stationary if and only if the rows of $\mathbf{A} - \mathbf{I}$ are cointegrating vectors.
- ▶ One possibility is that $(\mathbf{A} - \mathbf{I})$ has rank 0, in which case we do not have any cointegrating vectors.
- ▶ But if $(\mathbf{A} - \mathbf{I})\mathbf{X}_{t-1}$ is stationary and the rank of $\mathbf{A} - \mathbf{I}$ is $R \geq 1$ then there are R cointegrating vectors (linearly independent combinations of the components of stationary).

Cointegration: Johansen 3

- ▶ If $R = 0$, then there are no cointegrating vectors.
- ▶ If $1 \leq R \leq n - 1$, then there are R linearly independent cointegrating vectors.
- ▶ The Johansen method associates cointegration with determining the rank of $\mathbf{A} - \lambda \mathbf{I}$ or equivalently the number of eigenvalues of $\mathbf{A} - \lambda \mathbf{I}$ that are non-zero.
- ▶ These vectors are not uniquely determined, but they span the space of cointegrating vectors.

Cointegration: Johansen 4

The analysis for VAR(1) processes extends in a straightforward way to VAR(p) processes

$$\mathbf{X}_t = \mathbf{a} + \sum_{i=1}^p \mathbf{A}_i \mathbf{X}_{t-i} + \boldsymbol{\epsilon}_t.$$

- ▶ Again, we assume that $\mathbf{X}_t$ is $\mathcal{I}(1)$. If not, then the vector needs to be differenced enough times to bring it to $\mathcal{I}(1)$.
- ▶ Subtract $\mathbf{X}_{t-1}$ from both sides to find:

Cointegration: Johansen 5

$$\begin{aligned}\nabla \mathbf{X}_t &= \mathbf{a} + (\mathbf{A}_1 - \mathbf{I})\mathbf{X}_{t-1} + \sum_{i=2}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t \\ &= \mathbf{a} + (\mathbf{A}_1 - \mathbf{I})\nabla \mathbf{X}_{t-1} + (\mathbf{A}_1 + \mathbf{A}_2 - \mathbf{I})\mathbf{X}_{t-2} \\ &\quad + \sum_{i=3}^p \mathbf{A}_i \mathbf{X}_{t-i} + \epsilon_t\end{aligned}$$

- ▶ This continues sequentially.
- ▶ Let

$$\mathbf{B}_i = (\mathbf{A}_1 + \cdots + \mathbf{A}_i - \mathbf{I}),$$

then subtracting and adding $(\sum_{j=1}^k \mathbf{A}_j - \mathbf{I})$ on the right side yields:

Cointegration: Johansen 6

$$\nabla \mathbf{X}_t = \mathbf{a} + \sum_{j=1}^{p-1} \mathbf{B}_j \nabla \mathbf{X}_{t-j} + \mathbf{B}_p \mathbf{X}_{t-p} + \epsilon_t.$$

- ▶ Since we assumed $\mathbf{X}_t$ is $\mathcal{I}(1)$, all terms involving $\nabla \mathbf{X}_{t-j}$ are stationary, hence $\mathbf{B}_p \mathbf{X}_p$ must be stationary.
- ▶ The term $\mathbf{B}_p \mathbf{X}_{t-p}$ must be consistent with stationarity, so the arguments given for the $p = 1$ case applied to $(\mathbf{A} - \mathbf{I})$ now must be applied to $\mathbf{B}_p$.
- ▶ It's rank determines the number of cointegrating vectors, and the identity of those vectors.